Cambridge O Level Computer Science

Chapter 8: Programming

Lecture notes with Check-Yourself questions and a full answer key.

In this chapter you will learn about:

- Programming concepts: variables and constants, input and output, sequence, selection (including nesting), iteration, totalling and counting, string handling, arithmetic/logical/Boolean operators, procedures and functions (parameters, local and global variables), library routines, and creating maintainable programs.
- Arrays: one- and two-dimensional arrays, indexes, and using iteration to read from and write to arrays.
- File handling: opening, closing, reading from and writing to data and text files.

Chapter 7 taught you to design algorithms; this chapter turns them into programs. The programs are shown in pseudocode plus the three recommended IGCSE languages: Python (IDLE), Visual Basic (Visual Studio) and Java (BlueJ). The traditional first program displays 'Hello World' on the screen. This chapter covers the concepts; use it alongside learning your chosen language's syntax in an IDE.

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8.1 Programming concepts

Five basic constructs underlie every program: data use (variables, constants, arrays), selection (choosing a path), iteration (repeating steps), and operator use (arithmetic for calculations, logical and Boolean for decisions).

8.1.1 Variables and constants

A variable is a named data store holding a value that may change while the program runs; a constant is a named data store holding a value that does not change. Both need meaningful names so others can understand the program. Not all languages distinguish them explicitly, but the programmer must always know which stores can change. Constants are often highlighted, for example PI = 3.142, capital letters, or names beginning with Const (ConstPi = 3.142).

Good practice is to declare the constants and variables a program will use, either at the start or just before first use. Some languages need explicit declarations (stating

the data type); others use implicit declarations (the type comes from the assigned value).

How to declare variables and constants:

Pseudocode	Python	Visual Basic	Java
DECLARE FirstVar : INTEGER (explicit); CONSTANT FirstConst <- 100 (implicit)	No separate declarations; no distinction between constants and variables (the programmer must manage this)	Dim FirstVar As Integer (explicit); Const FirstConst As Integer = 500 (explicit, or implicit as Const First = 500)	int FirstVar; final int FIRSTCONST = 500; (constants are final variables, usually capitalised; variables often declared as used)

Check yourself

1. 1. What is the difference between a variable and a constant?
2. 2. Why should variables and constants be given meaningful names?
3. 3. Explain the difference between explicit and implicit declarations.
4. 4. Declare an integer variable called Score and a constant called MaxScore with value 100 in pseudocode.

8.1.2 Basic data types

Data types let the computer store and manipulate data effectively (numbers with mathematical operators, characters with concatenation) and allow automatic validation in some cases. The IGCSE data types:

- Integer: a positive or negative whole number, usable with mathematical operators.
- Real: a positive or negative number with a fractional part, usable with mathematical operators.
- Char: a single character.
- String: several characters (letters, digits, printable symbols); may be empty; a number stored as a string cannot be used in calculations.
- Boolean: TRUE or FALSE only.

Examples across languages:

Pseudocode	Python	Visual Basic	Java
INTEGER: FirstInteger <- 25	FirstInt = 25	Dim FirstInt As Integer	int FirstInt;

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REAL: FirstReal <- 25.0	FirstReal = 25.0	Dim FirstReal As Decimal	double FirstReal;
CHAR: Female <- 'F'	Female = 'F'	Dim Female As Char	char Female;
STRING: FirstName <- 'Emma'	FirstName = "Emma"	Dim FirstName As String	String FirstName;
BOOLEAN: Flag <- True	Flag = True	Dim Flag As Boolean	boolean Flag;

Check yourself

5. 1. Name the five basic data types and give an example value of each.
6. 2. Why can a number stored as a string not be used in calculations?
7. 3. For a program finding the volume of a cylinder, suggest suitable variables and constants with their data types.

8.1.3 Input and output

Programs take input from the keyboard and send output to the screen. Every input needs a prompt telling the user what to enter; every output needs a message explaining the result. Inputs default to strings, so converting to the required type is needed (casting): Python uses int() or float(). Parts of an output are joined by a separator character.

Input statements with prompts:

Pseudocode	Python	Visual Basic	Java
INPUT Radius (prompt shown separately)	radius = float(input("Please enter the radius of the cylinder ")) (prompt combined with input)	Console.Write("Please enter the radius of the cylinder ") then radius = Decimal.Parse(Console.ReadLine()) (separate prompt and input; input specifies the type)	Import Scanner, create input object; System.out.println("Please enter the radius of the cylinder "); double radius = myObj.nextDouble( ); (separate prompt and input; input specifies the type)

Output statements with messages (separators differ):

Python (comma)	Visual Basic (&)	Java (+)

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print("Volume of the cylinder is ", volume)	Console.WriteLine("Volume of the cylinder is " & volume)	System.out.println("Volume of the cylinder is " + volume);

Complete cylinder-volume program (Volume = Radius * Radius * Length * PI, with PI = 3.142):

```
CONSTPI = 3.142
Radius = float(input("Please enter the radius of the cylinder "))
Length = float(input("Please enter the length of the cylinder "))
Volume = Radius * Radius * Length * CONSTPI
print("Volume of the cylinder is ", Volume)
```

Every VB console program needs a Module with a Main procedure; every Java console program needs a class with a main procedure, and every Java statement ends with a semicolon.

Worked example: A cylinder has radius 3 and length 5. Calculate the volume using PI = 3.142.

Volume = Radius * Radius * Length * PI.

Volume = 3 * 3 * 5 * 3.142 = 9 * 5 * 3.142 = 45 * 3.142 = 141.39.

Check yourself

8. 1. Why does every input need a prompt and every output need a message?
9. 2. What is casting, and why is it needed for numeric input?
10. 3. State the output separator character used in Python, Visual Basic and Java.
11. 4. A cylinder has radius 2 and length 10. Calculate its volume using PI = 3.142.

8.1.4 Basic concepts

(a) Sequence

The order of steps matters: the wrong order gives wrong results or extra unnecessary steps. Example: an algorithm totalling marks until 999 is entered, also counting entries and averaging. The incorrect version calculates the average inside the loop (using the unfinished total), includes the 999 in the total and count, and tests for 999 unnecessarily. The corrected version: initialise Counter <- -1 (so the

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